

Electronic Supplementary Material for:
Distance functions of carabids in crop fields depend on functional traits, crop type and adjacent habitat: a synthesis

ESM 1: Supplementary tables and figures

Fabian A. Boetzel^{1,2*}, Douglas Sponsler², Matthias Albrecht³, Péter Batáry⁴, Klaus Birkhofer⁵, Michal Knapp⁶, Jochen Krauss², Bea Maas⁷, Emily A. Martin⁸, Clélia Sirami^{9,10}, Louis Sutter¹¹, Colette Bertrand^{12,13}, Alette Boser Baillod¹⁴, Gerard Bota¹⁵, Vincent Bretagnolle^{16,17}, Lluís Brotons^{15,18,19}, Thomas Frank²⁰, Moritz Fusser²¹, David Giralt¹⁵, Ezequiel González^{6,22}, Anouschka R. Hof²³, Henryk Luka¹⁴, Ronan Marrec²⁴, Michael A. Nash²⁵, Katherina Ng²⁶, Manuel Plantegenest²⁷, Brigitte Poulin²⁸, Gavin M. Siriwardena²⁹, Teja Tschardt³⁰, Matthias Tschumi^{3,31}, Aude Vialatte^{9,10}, Laura Van Vooren³², Muhammad Zubair-Anjum³³, Martin H. Entling²¹, Ingolf Steffan-Dewenter² & Jens Schirmer²¹

Affiliations:

¹ Swedish University of Agricultural Sciences, Department of Ecology, SE-750 07 Uppsala, Sweden

² Department of Animal Ecology and Tropical Biology, Biocenter, University of Würzburg, Am Hubland, 97074 Würzburg, Germany

³ Agroecology and Environment, Agroscope, Reckenholzstrasse 191, 8046 Zurich, Switzerland

⁴ “Lendület” Landscape and Conservation Ecology, Institute of Ecology and Botany, HUN-REN Centre for Ecological Research, 2163 Vácrátót, Alkotmány út 2-4, Hungary

⁵ Department of Ecology, Brandenburg University of Technology Cottbus-Senftenberg, 03046 Cottbus, Germany

⁶ Faculty of Environmental Sciences, Czech University of Life Sciences Prague, Kamýcká 129, Praha-Suchbát, 165 00, Czech Republic

- ⁷ Department of Botany and Biodiversity Research, Division of Biodiversity Dynamics and Conservation, University of Vienna, Rennweg 14, 1030 Vienna, Austria
- ⁸ Department of Animal Ecology, Justus Liebig University Giessen, Heinrich-Buff-Ring 26-32, 35392 Giessen, Germany
- ⁹ UMR Dynafor, INRAE, Toulouse University, 31326 Castanet Tolosan, France
- ¹⁰ LTSER Zone Atelier « PYRÉNÉES GARONNE », 31320 Auzerville-Tolosane, France
- ¹¹ Plant-Production Systems, Agroscope, Route des Eterpys 18, 1964 Conthey, Switzerland
- ¹² Université Paris-Saclay, INRAE, AgroParisTech, UMR EcoSys, 91120 Palaiseau, France
- ¹³ INRAE, Institut Agro, ESA, UMR BAGAP, 35042, Rennes, France
- ¹⁴ Research Institute of Organic Agriculture (FiBL), Ackerstrasse 113, Postfach, Frick 5070, Switzerland
- ¹⁵ Landscape Dynamics and Biodiversity Program, Forest Science and Technology Centre of Catalonia (CTFC), Crtra. Sant Llorenç de Morunys, km 2, 25280 Solsona, Spain.
- ¹⁶ CEBC, UMR 7372, CNRS & La Rochelle Université, 79360 Villiers-en-Bois, France
- ¹⁷ LTSER “Zone Atelier Plaine & Val de Sèvre”, CNRS, 79360 Villiers-en-Bois, France
- ¹⁸ CREAM, Cerdanyola del Vallès, 08193, Spain
- ¹⁹ CSIC, Cerdanyola del Vallès, 08193, Spain
- ²⁰ Institute of Zoology, University of Natural Resources and Life Sciences, A-1180 Vienna, Austria
- ²¹ University of Kaiserslautern-Landau, iES Landau, Institute for Environmental Sciences, Ecosystem Analysis, Fortstrasse 7, Landau 76829, Germany
- ²² Instituto Multidisciplinario de Biología Vegetal (CONICET-Universidad Nacional de Córdoba), Av. Velez Sarsfield 1611, 5000, Córdoba, Argentina
- ²³ Wildlife Ecology and Conservation Group, Wageningen University, Droevendaalsesteeg 3, 6708 PB, Wageningen, the Netherlands

²⁴ 2FRE CNRS 3498 EDYSAN (Écologie et Dynamique des Systèmes Anthropisés), Université de Picardie Jules Verne, UFR des Sciences – Batiment des Minimes, 2 rue Edmond Fontaine, 80000 Amiens, France

²⁵ Department of Ecology, Environment & Evolution, School of Life Science, La Trobe University, Bundoora, VIC 3086

²⁶ Fenner School of Environment and Society, The Australian National University, Canberra, Australia

²⁷ Institut Agro, Univ Rennes1, INRAE, IGEPP, 35000 Rennes, France.

²⁸ Tour du Valat Research Institute for the conservation of Mediterranean wetlands, Le Sambuc, 13200 Arles, France

²⁹ British Trust for Ornithology, The Nunnery, Thetford, Norfolk IP24 2PU, UK

³⁰ Agroecology, Department of Crop Science, University of Göttingen, Göttingen, Germany

³¹ Swiss Ornithological Institute, Seerose 1, CH-6204 Sempach, Switzerland

³² Ghent University, Faculty of Bioscience Engineering, Department of Forest and Water Management, Forest & Nature Lab, Geraardsbergsesteenweg 267, 9090 Gontrode

³³ Department of Zoology & Biology, Faculty of Sciences, Pir Mehr Ali Shah Arid Agriculture University Rawalpindi, Rawalpindi, Pakistan

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Search terms

Search terms used in the 'Web of Science' search (initial search conducted on 06.11.2018):

carabid* OR "ground beetle"

AND

"semi-natural habitat" OR "seminatural habitat" OR margin* OR
boundary OR strip OR "buffer strip" OR shelterbelt OR hedge* OR
border OR "field border" OR fringe OR "agri-environmental scheme" OR
"within-field distribution" OR "distance to edge" OR "field edge" OR
"field distribution" OR "spill-over" OR spillover

AND

"cropping system" OR field* OR crop* OR farmland OR agriculture OR
wheat OR arable OR maize OR "oilseed rape" OR Canola

AND

diversity OR "species richness" OR biodiversity OR abundance OR
density

Figure S1 Reported distance functions in the literature

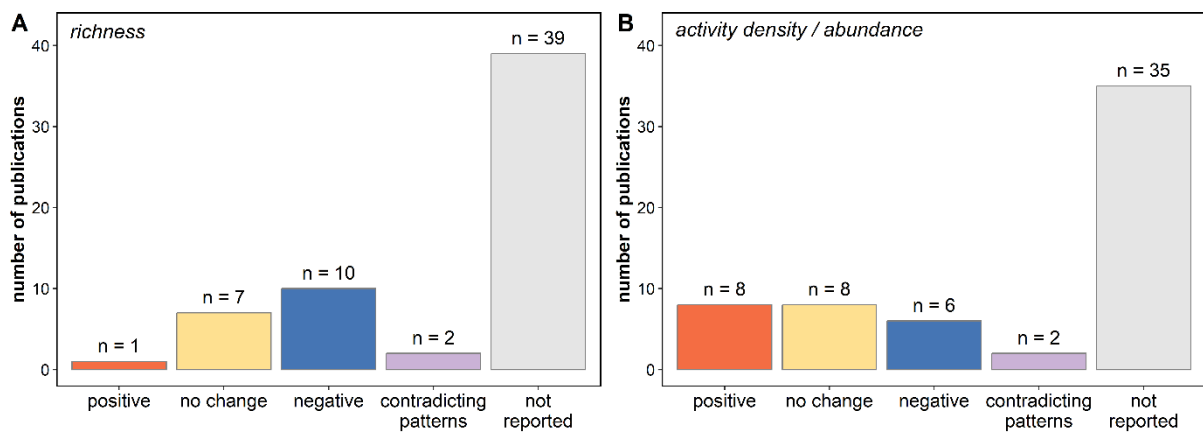


FIGURE S1: Reported distance functions for carabid richness (A) and carabid activity density or abundance (B) in the publications detected as potentially suitable based on our literature search (44 publications) and by asking colleagues (15 publications). ‘Positive’ (red) and ‘negative’ (blue) indicate a significant increase or decrease reported in the publication, ‘no change’ (yellow) indicates no significant distance function was found, ‘contradicting patterns’ (violet) indicates that distance functions differed depending on the context within the same publication. For both richness and activity density, a majority of the publications did not report distance functions despite data being collected with a suitable design (‘not reported’; grey).

Figure S2 Map of crop types and adjacent habitat types in the datasets

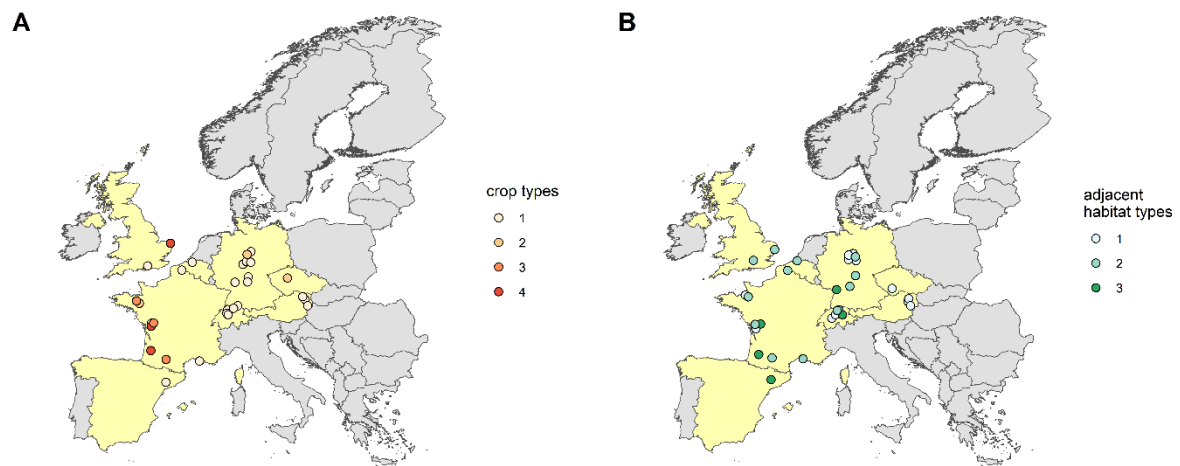


FIGURE S2: Map of European countries from which datasets were used for the analyses (yellow colouration). Colouration of the points indicates the number of crop types (A) and adjacent habitat types in each dataset, (B). Points jittered by 0.5° to reduce overlay. For more information on the datasets, see Table S2.

Figure S3 Effect of trap days on carabid richness.

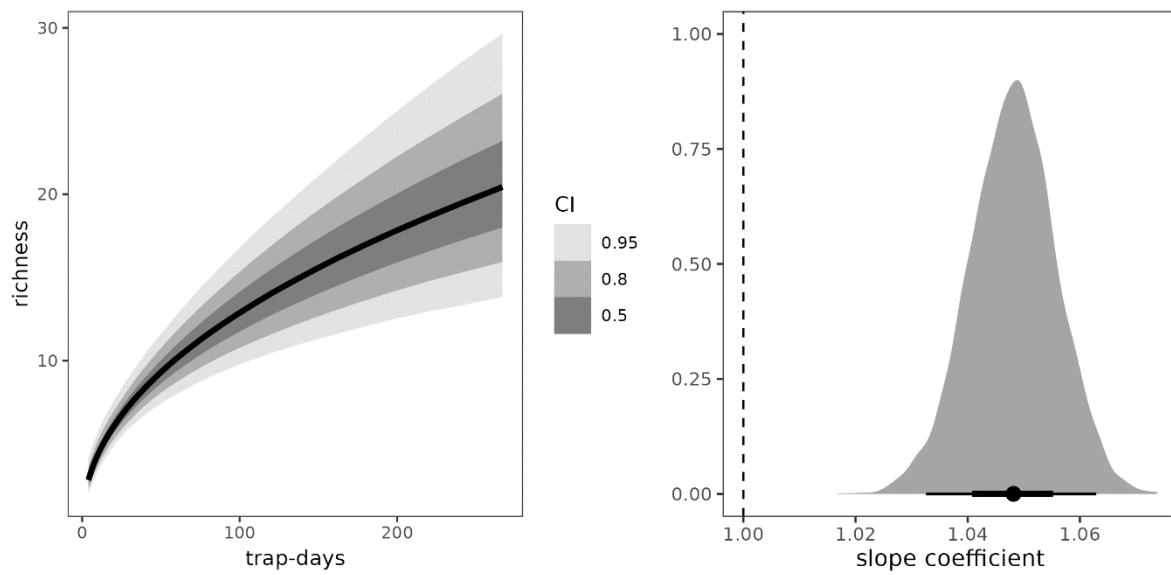


FIGURE S3: Marginal effect of carabid beetle richness across trap days (traps pooled per plot, see materials and methods; prediction derived from the model, see Table S5). The response curve on the left depicts predicted richness with 95% credible interval (CI). The density plot on the right shows the slope coefficient (on the log scale) of trap days with median (point) 66% (thick bars) and 95% (thin bars) posterior credible intervals. The value of 1, indicated with a vertical dashed line, corresponds to a perfectly flat slope, and parameters with 95% credible intervals falling entirely above or below 1 can be regarded as positive or negative slopes, respectively. The rug in the left panel indicates numbers of days for which data points were available, predictions are limited to the range of trapping days covered by the available data.

Figure S4 Overall predictions for omnivorous species

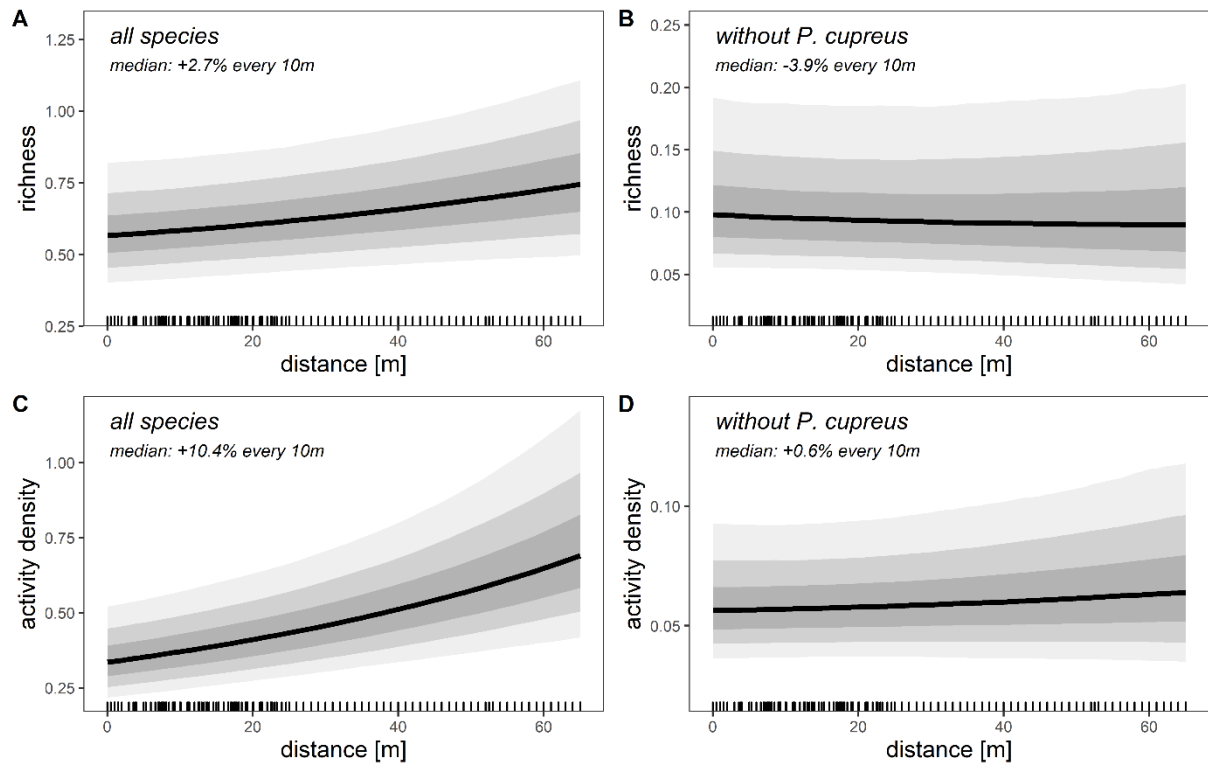


FIGURE S4: Overall marginal predictions for (A & B) the richness of omnivorous carabid species and (C & D) the activity density of omnivorous carabid species across the within field distance gradient both with all species included (A & C) and with the hyperdominant species *Poecilus cupreus* removed (B & D). Response curves depict posterior predictions with 50% (dark), 80% (medium) and 95% (light) credible intervals (CI), back-transformed from the log-link scale to response scale. The rug indicates distances for which data points were available, predictions are limited to the distances covered by the available data (not all data used in the models is shown; distance was clipped at 65 m for visualization as a majority of the data was available up to this distance). For statistics and values, see electronic supplementary material 1: tables S8 & S9.

Figure S5 Predictions for omnivorous species in relation to crop and adjacent habitat type

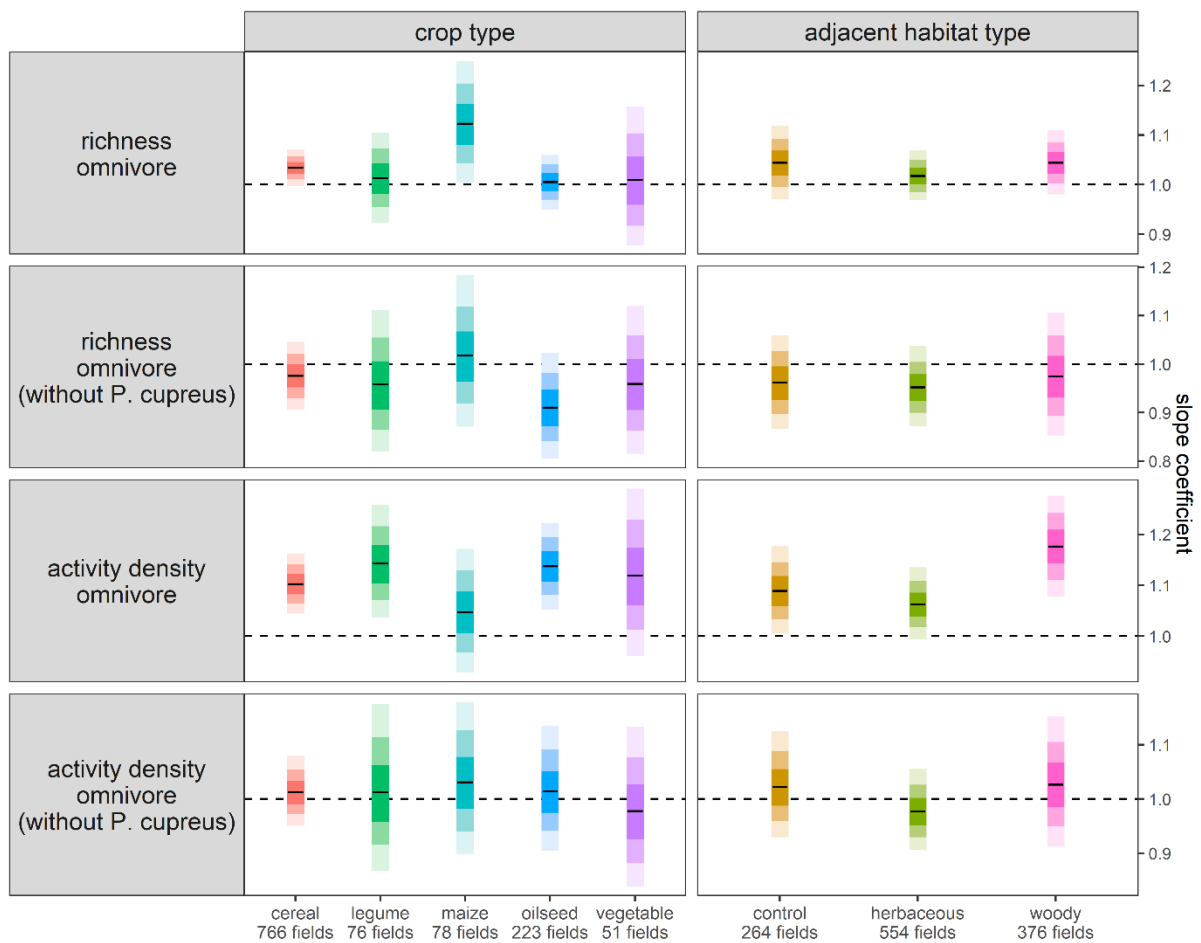


FIGURE S5: Overall marginal within-field distance slope coefficients for the richness and activity density of omnivorous carabid species in relation to crop type (left) and adjacent habitat type (right) both with all species included and with the hyperdominant species *Poecilus cupreus* removed. Slope coefficient estimates (median) of the posterior sample (black lines) with 50% (dark), 80% (medium) and 95% (light) credible intervals (CI), back-transformed from the log link scale to response scale. The dashed line represents the slope coefficient of 1, i.e. a flat slope. Coefficients can be interpreted as percent change in the response every 10 m towards the field centre. For statistics and values, see electronic supplementary material 1: tables S8 & S9.

Table S1 Publications returned by the literature search

Table S1: List papers obtained from the initial literature search in the ISI Web of Science Core Collection and deemed suitable after full text review. For all papers, data were requested from the authors. For exclusion reasons for the datasets received but ultimately not included, see exclusion criteria stated in the text and Table S3.

Reference	data received after request	data included after review
Al Hassan, D; Georgelin, E; Delattre, T; Burel, F; Plantegenest, M; Kindlmann, P; Butet, A (2013) Does the presence of grassy strips and landscape grain affect the spatial distribution of aphids and their carabid predators? <i>Agricultural and Forest Entomology</i> 15, 24-33.		
Anjum-Zubair, M; Schmidt-Entling, MH; Querner, P; Frank, T (2010) Influence of within-field position and adjoining habitat on carabid beetle assemblages in winter wheat. <i>Agricultural and Forest Entomology</i> 12, 301-306.	X	X
Bedford, SE; Usher, MB (1994) Distribution of arthropod species across the margins of farm woodlands. <i>Agriculture, Ecosystems & Environment</i> 48, 295-305.		
Birkhofer, K; Fevrier, V; Heinrich, AE; Rink, K; Smith, HG (2018) The contribution of CAP greening measures to conservation biological control at two spatial scales. <i>Agriculture, Ecosystems & Environment</i> 255, 84-94.		
Birkhofer, K; Wolters, V; Diekotter, T (2014) Grassy margins along organically managed cereal fields foster trait diversity and taxonomic distinctness of arthropod communities. <i>Insect Conservation and Diversity</i> 7, 274-287.	X	X
Chapelin-Viscardi, JD; Maillat-Mezeray, J; Tosser, V; Wartelle, R (2014) Emergence of Ground beetles in farming areas. Choice of habitat, diversity and specific requirements (Carabidae, Coleoptera). <i>Bulletin Mensuel de la Société Linnéenne de Lyon</i> 83, 157-170.		
Cividanes, FJ; Araujo, ES; Ide, S; Galli, JC (2010) Distribution and habitat preference of Carabidae and Staphylinidae (Coleoptera) in an orange orchard and forest fragment. <i>Florida Entomologist</i> 93, 339-345.		
Cividanes, FJ; dos Santos-Cividanes, TM; Ferraudo, AS; da Matta, DH (2018) Edge effects on carabid beetles (Coleoptera: Carabidae) between forest fragments and agricultural fields in south-east Brazil. <i>Austral Entomology</i> 57, 9-16.		
Collins, KL; Boatman, ND; Wilcox, A; Holland, JM; Chaney, K (2002) Influence of beetle banks on cereal, aphid predation in winter wheat. <i>Agriculture, Ecosystems & Environment</i> 93, 337-350.		
Dennis, P; Fry, GLA (1992) Field margins – can they enhance natural enemy population-densities and general arthropod diversity on farmland? <i>Agriculture, Ecosystems & Environment</i> 40, 95-115.		
Ditner, N; Balmer, O; Beck, J; Blick, T; Nagel, P; Luka, H (2013) Effects of experimentally planting non-crop flowers into cabbage fields on the abundance and diversity of predators. <i>Biodiversity and Conservation</i> 22, 1049-1061.	X	X
Ernault, A; Vialatte, A; Butet, A; Michel, N; Rantier, Y; Jambon, O; Burel, F (2013) Grassy strips in their landscape context, their role as new habitat for biodiversity. <i>Agriculture, Ecosystems & Environment</i> 166, 15-27.		
Fernandes M, Ivan C; Cividanes, FJ; Ide, S; Haddad, GQ (2012) Diversity and habitat preferences of Carabidae and Staphylinidae (Coleoptera) in two agroecosystems. <i>Bragantia</i> 71.		
Ferrante, M; Gonzalez, E; Lövei, GL (2017) Predators do not spill over from forest fragments to maize fields in a landscape mosaic in central Argentina. <i>Ecology and Evolution</i> 7, 7699-7707.	X	
Fournier, E; Loreau, M (1999) Effects of newly planted hedges on ground-beetle diversity (Coleoptera, Carabidae) in an agricultural landscape. <i>Ecography</i> 22, 87-97.		
Fournier, E; Loreau, M (2002) Foraging activity of the carabid beetle <i>Pterostichus melanarius</i> Ill. in field margin habitats. <i>Agriculture, Ecosystems & Environment</i> 89, 253-259.		

Frank, T (1997) Species diversity of ground beetles (Carabidae) in sown weed strips and adjacent fields. <i>Biological Agriculture & Horticulture</i> 15, 297-307.		
French, BW; Elliott, NC (1999) Temporal and spatial distribution of ground beetle (Coleoptera : Carabidae) assemblages in grasslands and adjacent wheat fields. <i>Pedobiologia</i> 43, 73-84.	X (2 datasets)	
García, AF; Griffiths, GJK; Thomas, CFG (2000) Density, distribution and dispersal of the carabid beetle <i>Nebria brevicollis</i> in two adjacent cereal fields. <i>Annals of Applied Biology</i> 137, 89-97.		
Hassal, M; Hawthorne, A; Maudsley, M; White, P; Cardwell, C (1992) Effects of headland management on invertebrate communities in cereal fields. <i>Agriculture, Ecosystems & Environment</i> 40, 155-178.		
Hawthorne, AJ; Hassall, M; Sotherton, NW (1998) Effects of cereal headland treatments on the abundance and movements of three species of carabid beetles. <i>Applied Soil Ecology</i> 9, 417-422.		
Hof, AR; Bright, PW (2010) The impact of grassy field margins on macro-invertebrate abundance in adjacent arable fields. <i>Agriculture, Ecosystems & Environment</i> 139, 280-283.	X	X
Kagawa, Y; Maeto, K (2009) Spatial population structure of the predatory ground beetle <i>Carabus yaconinus</i> (Coleoptera: Carabidae) in the mixed farmland-woodland satoyama landscape of Japan. <i>European Journal of Entomology</i> 106, 385-391.		
Kagawa, Y; Maeto, K (2014) Ground beetle (Coleoptera: Carabidae) assemblages associated with a satoyama landscape in Japan: the effects of soil moisture, weed height, and distance from woodlands. <i>Applied Entomology and Zoology</i> 49, 429-436.		
Kajak, A; Oleszczuk, M (2004) Effect of shelterbelts on adjoining cultivated fields: Patrolling intensity of carabid beetles (Carabidae) and spiders (Araneae). <i>Polish Journal of Ecology</i> 52, 155-172.		
Kujawa, K; Sobczyk, D; Kajak, A (2006) Dispersal of <i>Harpalus rufipes</i> (Degeer)(Carabidae) between shelterbelt and cereal field. <i>Polish Journal of Ecology</i> 54, 243-252.	X	
Labruyere, Sarah; Petit, Sandrine; Ricci, Benoit (2018) Annual variation of oilseed rape habitat quality and role of grassy field margins for seed eating carabids in arable mosaics. <i>Agricultural and Forest Entomology</i> 20, 234-245.		
Lee, JC; Menalled, FB; Landis, DA (2001) Refuge habitats modify impact of insecticide disturbance on carabid beetle communities. <i>Journal of Applied Ecology</i> 38, 472-483.		
Lys, JA; Zimmermann, M; Nentwig, W (1994) Increase in activity density and species number of carabid beetles in cereals as a result of strip-management. <i>Entomologia Experimentalis et Applicata</i> 73, 1-9.		
Madeira, F; Tschamtké, T; Elek, Z; Kormann, UG; Pons, X; Rosch, V; Samu, F; Scherber, C; Batary, P (2016) Spillover of arthropods from cropland to protected calcareous grassland - the neighbouring habitat matters. <i>Agriculture, Ecosystems & Environment</i> 235, 127-133.	X	X
Marasas, ME; Sarandon, SJ; Cicchino, A (2010) Semi-Natural Habitats and Field Margins in a Typical Agroecosystem of the Argentinean Pampas as a Reservoir of Carabid Beetles. <i>Journal of Sustainable Agriculture</i> 34, 153-168.		
Marrec, R; Badenhäusser, I; Bretagnolle, V; Borger, L; Roncoroni, M; Guillon, N; Gauffre, B (2015) Crop succession and habitat preferences drive the distribution and abundance of carabid beetles in an agricultural landscape. <i>Agriculture, Ecosystems & Environment</i> 199, 282-289.	X (3 datasets)	X (3 datasets)
Nash, MA; Thomson, LJ; Hoffmann, AA (2008) Effect of remnant vegetation, pesticides, and farm management on abundance of the beneficial predator <i>Notonomus gravis</i> (Chaudoir) (Coleoptera : Carabidae). <i>Biological Control</i> 46, 83-93.	X	
Ng, K; Barton, PS; Macfadyen, S; Lindenmayer, DB; Driscoll, DA (2018) Beetle's responses to edges in fragmented landscapes are driven by adjacent farmland use, season and cross-habitat movement. <i>Landscape Ecology</i> 33, 109-125.	X	
Outward, R; Sorenson, CE; Bradley, JR (2008) Effects of vegetated field borders on arthropods in cotton fields in eastern North Carolina. <i>Journal of Insect Science</i> 8.		

Rouabah, A; Villerd, J; Amiaud, B; Plantureux, S; Lasserre-Joulin, F (2015) Response of carabid beetles diversity and size distribution to the vegetation structure within differently managed field margins. <i>Agriculture, Ecosystems & Environment</i> 200, 21-32.		
Roume, A; Ouin, A; Raison, L; Deconchat, M (2011) Abundance and species richness of overwintering ground beetles (Coleoptera: Carabidae) are higher in the edge than in the centre of a woodlot. <i>European Journal of Entomology</i> 108, 615-622.		
Saska, P; Vodde, M; Heijerman, T; Westerman, P; van der Werf, W (2007) The significance of a grassy field boundary for the spatial distribution of carabids within two cereal fields. <i>Agriculture, Ecosystems & Environment</i> 122, 427-434.	X	
Soboleva-Dokuchaeva, II; Chernyshev, VB; Afonina, VM; Ovchinnikova, MF; Timokhov, AV (2000) Factors responsible for distribution of ground beetles (Coleoptera, Carabidae) in agricultural lands. <i>Zoologicheskii Zhurnal</i> 79, 1071-1072.		
Soboleva-Dokuchaeva, II; Tshernyshev, VB; Afonina, VM; Timokhov, AV (2000) Seasonal dynamics of spatial distribution of mass ground-beetle species (Coleoptera, Carabidae) in agroecosystems of the mixed forest zone. <i>Zoologicheskii Zhurnal</i> 79, 822-823.		
Sutter, L; Amato, M; Jeanneret, P; Albrecht, M (2018) Overwintering of pollen beetles and their predators in oilseed rape and semi-natural habitats. <i>Agriculture, Ecosystems & Environment</i> 265, 275-281.	X	X
Thomas, CFG; Green, F; Marshall, EJP (1997) Distribution, dispersal and population size of the ground beetles, <i>Pterostichus melanarius</i> (Illiger) and <i>Harpalus rufipes</i> (Degeer) (Coleoptera, carabidae), in field margin habitats. <i>Biological Agriculture & Horticulture</i> 15, 337-352.		
Thomas, CFG; Marshall, EJP (1999) Arthropod abundance and diversity in differently vegetated margins of arable fields. <i>Agriculture, Ecosystems & Environment</i> 72, 131-144.		
Van Vooren, L; Reubens, B; Ampoorter, E; Broekx, S; Pardon, P; Van Waes, C; Verheyen, K (2018) Monitoring the Impact of Hedgerows and Grass Strips on the Performance of Multiple Ecosystem Service Indicators. <i>Environmental Management</i> 62, 241-259.	X	X
Total	18	11

Table S2 Datasets included in the analyses

Table S2: List of all used datasets with country, region, study years, crop types, adjacent habitat types, study sites (after filtering), maximal sampled distances [m] and sampling days included in the datasets.										
Study ID	country	region	year	months	crop types	adjacent habitat classes	study sites	maximal distance [m]	trapping days	reference
Bata01	Germany	Lower Saxony	2011	July / August	cereals	herbaceous	7	10	14	Madeira <i>et al.</i> (2016)
Bata02	Germany	Lower Saxony	2008	June	cereals	herbaceous	18	30	35	Batáry <i>et al.</i> (2012)
Bata03	Germany	Lower Saxony	2013	June	cereals	herbaceous	36	200	28	Batáry <i>et al.</i> (2017)
Birk01	Germany	Lower Saxony	2008	June / July	cereals	herbaceous	17	40	14	Birkhofer, Wolters and Diekötter (2014)
Boet01	Germany	Lower Frankonia	2016	April – July	oilseed crops	herbaceous, control	31	65	252	Boetzl <i>et al.</i> (2019)
Boet02	Germany	Lower Frankonia	2017	April – July	cereals	herbaceous, control	30	65	252	Boetzl <i>et al.</i> (2020)
Entl01	Switzerland	Solothurn	2005	May / June	cereals	herbaceous	20	30	48	Anjum-Zubair <i>et al.</i> (2010)
Farm01	France	Armorique	2013 / 2014	May – July	cereals, maize	herbaceous, woody	104	25	16	Sirami <i>et al.</i> (2019)
Farm02	France	Camargue	2013 / 2014	May – July	cereals	herbaceous, woody	17	25	16	Sirami <i>et al.</i> (2019)
Farm03	France	Coteaux	2013 / 2014	May – July	cereals, maize, leguminous crops, oilseed crops	herbaceous, woody, control	84	25	16	Sirami <i>et al.</i> (2019)
Farm04	UK	EastAnglia	2012 / 2013	May – July	cereals, maize, oilseed crops, vegetables	herbaceous, woody	150	25	16	Sirami <i>et al.</i> (2019)
Farm05	Germany	Lower Saxony	2013 / 2014	May – July	cereals, oilseed crops	herbaceous, control	154	25	16	Sirami <i>et al.</i> (2019)
Farm06	Spain	Catalonia	2013 / 2014	April – June	cereals	herbaceous, woody, control	34	25	16	Sirami <i>et al.</i> (2019)
Farm08	France	Nouvelle-Aquitaine	2013 / 2014	May – July	cereals, maize, leguminous crops, oilseed crops	herbaceous, woody, control	249	25	16	Sirami <i>et al.</i> (2019)
Hof01	UK	Berkshire	2009	May – July	cereals	herbaceous, woody	32	20	15	Hof and Bright (2010)
Knap01	Czech Republic	Central Bohemia	2013	June / July	cereals, oilseed crops	herbaceous, woody	4	20	114	Knapp <i>et al.</i> (2019)
Luka01	Switzerland	Aargau	2010	June – September	vegetables	herbaceous	7	20	70	Ditner <i>et al.</i> (2013)
Luka02	Switzerland	Aargau	1996 – 1998	May – July	cereals	herbaceous	3	70	280	Pfiffner and Luka (2003)
Maas01	Austria	Lower Austria	2017	April / May	cereals	herbaceous	5	157.5	21	Maas <i>et al.</i> (2021)
Maas02	Austria	Lower Austria	2018	April / May	cereals	herbaceous	5	157.5	21	Maas <i>et al.</i> (2021)
Maas03	Austria	Lower Austria	2019	April / May	cereals	herbaceous	5	157.5	21	Maas <i>et al.</i> (2021)
Marr01	France	Nouvelle-Aquitaine	2011 – 2013	April – July	cereals, leguminous crops, oilseed crops	herbaceous, woody	38	94	18	Marrec <i>et al.</i> (2017)
Marr02	France	Armorique	2010 / 2011	April – July	cereals, leguminous crops, oilseed crops	herbaceous, woody	49	143	28	Marrec <i>et al.</i> (2017)
Marr03	France	Gascogne	2010 – 2012	April – July	cereals, leguminous crops, oilseed crops	herbaceous, woody	37	97	11	Marrec <i>et al.</i> (2017)
Schi01	Germany	Palatinate	2014	May / June	vegetables	herbaceous, woody, control	17	26	14	Fusser <i>et al.</i> (2018)
Sutt01	Switzerland	Zuerich	2014	April / May	oilseed crops	herbaceous, woody, control	18	25	28	Sutter <i>et al.</i> (2018)
Tsch01	Switzerland	Zuerich	2012	May – July	cereals	herbaceous, control	25	25	42	Tschumi <i>et al.</i> (2015)
Voor01	Belgium	North Flanders & Heuveland	2014	May / June	cereals	herbaceous, woody	8	30	42	Van Vooren <i>et al.</i> (2018)

Table S3 Datasets excluded from the analyses

Table S3: List of all datasets excluded from our analyses with country, region, study years, crop types, adjacent habitat types, study sites, maximal sampled distances [m] and sampling days included in the datasets. Exclusion reason is highlighted in red. Datasets marked with ‘*’ do not contain species level IDs.

Study ID	country	region	year	months	crop types	adjacent habitat classes	study sites	maximal distance [m]	trapping days (per trap)	reference
Bade01	France	Nouvelle-Aquitaine	2015	July	oilseed crops	herbaceous	23	45	4	Badenhausser <i>et al.</i> (2020)
Cutl01*	Canada	Nova Scotia	2008 / 2009	May – August	blueberry	woody	4	25	24 – 36	Cutler <i>et al.</i> (2012)
Cutl02*	Canada	Nova Scotia	2017	‘summer’	blueberry	woody	6	160	21	Loureiro A (2018)
Farm07	Canada	Ontario	2013 / 2014	April – June	cereals, leguminous crops, vegetables	herbaceous, woody, control	238	25	16	Sirami <i>et al.</i> (2019)
Fren01	USA	Oklahoma	1993 - 1996	various	cereals	herbaceous	6	50	203 – 237	French <i>et al.</i> (1999)
Fren02	USA	Oklahoma	1995 / 1996	various	cereals	herbaceous	9	50	68 – 410	French <i>et al.</i> (2001)
Gonz01	Argentina	Cordoba	2011 / 2012	December - February	leguminous crops	woody	9	100	7	González <i>et al.</i> (2017)
Juch01	Poland	Zachodniopomorskie	2018	June – September	vegetables	herbaceous	1	29	69	?
Koiv01	Finland	Kanta-Häme	2001	May – August	vegetables	woody	2	30	66	Koivula <i>et al.</i> (2004)
Mart01	South Korea	Gangwon-do	2009	June – September	leguminous crops, vegetables	herbaceous	28	20	20	Martin <i>et al.</i> (2016)
Nash01	Australia	Victoria	2006	February / March	forage grass	herbaceous	20	200	32	Nash <i>et al.</i> (2008)
Ng01	Australia	New South Wales	2014 / 2015	September / October January / February	cereals	woody	11	200	28	Ng <i>et al.</i> (2018)
Sask01	Netherlands	Gelderland	2004	March – July	cereals	herbaceous	2	49	146	Saska <i>et al.</i> (2007)

Table S4 Study sites per crop type and adjacent habitat.

Table S4: Study sites per crop type and adjacent habitat type.		
Crop type	Adjacent habitat type	Number of sites
cereals	none	160
	herbaceous	399
	woody	207
maize	none	12
	herbaceous	8
	woody	58
oilseed crops	none	62
	herbaceous	109
	woody	52
vegetables	none	7
	herbaceous	22
	woody	22
leguminous crops	none	23
	herbaceous	16
	woody	37

Table S5 Carabid beetle species and their traits.

Table S5: The 241 carabid species contained in the accumulated dataset with total number of individuals, number of study sites on which the species occur as well as their (predominant) diet and mean body size. Carabids not identified to species level were excluded in this table. Carabid beetle traits were obtained from the database carabids.org (Homburg et al. 2014), monographies as well as previous studies on carabid beetle functional traits (Hůrka 1996; Ribera et al. 2001; Pakeman & Stockan 2014; Hanson et al. 2016).

Species	total individuals	number of sampling plots with occurrence	diet	mean body size [mm]
<i>Abax ovalis</i>	5	4	carnivorous	13
<i>Abax parallelepipedus</i>	81	58	carnivorous	19
<i>Abax parallelus</i>	37	19	carnivorous	15.5
<i>Acupalpus elegans</i>	9	4	omnivorous	3.5
<i>Acupalpus interstitialis</i>	5	5	omnivorous	2.5
<i>Acupalpus meridianus</i>	354	150	omnivorous	3.5
<i>Agonum emarginatum</i>	7	7	carnivorous	8
<i>Agonum gracilipes</i>	1	1	carnivorous	7.5
<i>Agonum lugens</i>	6	5	carnivorous	8.5
<i>Agonum micans</i>	4	3	carnivorous	6
<i>Agonum muelleri</i>	4982	462	carnivorous	7.5
<i>Agonum nigrum</i>	10	8	carnivorous	7
<i>Agonum sexpunctatum</i>	200	132	carnivorous	7.5
<i>Agonum viduum</i>	10	8	carnivorous	8
<i>Agonum viridicupreum</i>	5	2	carnivorous	8.5
<i>Amara aenea</i>	741	381	granivorous	7
<i>Amara anthobia</i>	2	2	granivorous	6
<i>Amara apricaria</i>	24	18	granivorous	7.5
<i>Amara aulica</i>	87	33	granivorous	12.5
<i>Amara bifrons</i>	59	28	granivorous	6
<i>Amara communis</i>	37	26	granivorous	6
<i>Amara consularis</i>	347	79	granivorous	8
<i>Amara convexior</i>	78	43	granivorous	8
<i>Amara eurynota</i>	283	73	granivorous	11
<i>Amara familiaris</i>	487	185	granivorous	7
<i>Amara fulva</i>	6	4	granivorous	9
<i>Amara littorea</i>	35	22	granivorous	7.5
<i>Amara lucida</i>	4	3	granivorous	5
<i>Amara lunicollis</i>	74	56	granivorous	7.5
<i>Amara montivaga</i>	29	20	granivorous	8
<i>Amara nitida</i>	1	1	granivorous	8
<i>Amara ovata</i>	17049	470	granivorous	8.5
<i>Amara plebeja</i>	308	127	granivorous	7
<i>Amara rufipes</i>	4	4	granivorous	8.5
<i>Amara sabulosa</i>	2	2	granivorous	5.5
<i>Amara similata</i>	4417	438	granivorous	8.5
<i>Amara tricuspidata</i>	2	2	granivorous	7.5
<i>Amblystomus niger</i>	1	1	granivorous	2.5
<i>Anchomenus dorsalis</i>	78281	1816	carnivorous	6.5
<i>Anisodactylus binotatus</i>	544	189	granivorous	10.5
<i>Anisodactylus poeciloides</i>	1	1	granivorous	11.5

<i>Anisodactylus signatus</i>	46	25	omnivorous	12
<i>Asaphidion flavipes</i>	698	330	carnivorous	3.5
<i>Asaphidion stierlini</i>	14	11	carnivorous	3.5
<i>Badister bullatus</i>	434	161	carnivorous	5
<i>Badister lacertosus</i>	40	18	carnivorous	6.5
<i>Badister meridionalis</i>	2	2	carnivorous	6.5
<i>Badister sodalis</i>	519	123	carnivorous	3.5
<i>Badister unipustulatus</i>	2	2	carnivorous	8
<i>Bembidion biguttatum</i>	42	8	carnivorous	3.5
<i>Bembidion deletum</i>	1	1	carnivorous	4.5
<i>Bembidion femoratum</i>	63	17	carnivorous	4.5
<i>Bembidion guttula</i>	129	32	carnivorous	2.5
<i>Bembidion iricolor</i>	11	5	carnivorous	4
<i>Bembidion lampros</i>	9074	1192	carnivorous	3
<i>Bembidion latinum</i>	1	1	carnivorous	5
<i>Bembidion lunulatum</i>	1174	78	carnivorous	3.5
<i>Bembidion mannerheimii</i>	1	1	carnivorous	2.5
<i>Bembidion normannum</i>	1	1	carnivorous	2.5
<i>Bembidion obtusum</i>	7700	606	carnivorous	2.5
<i>Bembidion properans</i>	872	161	carnivorous	3.5
<i>Bembidion quadrimaculatum</i>	4767	211	carnivorous	2.5
<i>Bembidion tetracolum</i>	1839	399	carnivorous	5
<i>Blemus discus</i>	14	6	carnivorous	4.5
<i>Brachinus crepitans</i>	21921	353	carnivorous	8
<i>Brachinus elegans</i>	30	18	carnivorous	7.5
<i>Brachinus explodens</i>	1079	174	carnivorous	5.5
<i>Brachinus immaculicornis</i>	14	11	carnivorous	8.5
<i>Brachinus sclopetata</i>	2637	195	carnivorous	5.5
<i>Bradycellus caucasicus</i>	1	1	omnivorous	3.5
<i>Bradycellus harpalinus</i>	3	2	omnivorous	4
<i>Bradycellus verbasci</i>	5	96	omnivorous	4.5
<i>Broscus cephalotes</i>	34	7	granivorous	21
<i>Calathus ambiguus</i>	80	17	carnivorous	10
<i>Calathus cinctus</i>	158	9	carnivorous	7.5
<i>Calathus erratus</i>	8	6	carnivorous	10
<i>Calathus fuscipes</i>	4872	267	carnivorous	12
<i>Calathus melanocephalus</i>	44	25	carnivorous	7.5
<i>Calathus rotundicollis</i>	1	1	carnivorous	9.5
<i>Callistus lunatus</i>	1	1	carnivorous	5.5
<i>Calosoma inquisitor</i>	4	99	carnivorous	21.5
<i>Calosoma maderae</i>	7	5	carnivorous	27.5
<i>Carabus auratus</i>	1326	251	carnivorous	23.5
<i>Carabus auronitens</i>	4	4	carnivorous	25.5
<i>Carabus cancellatus</i>	189	90	carnivorous	26
<i>Carabus convexus</i>	33	19	carnivorous	17
<i>Carabus coriaceus</i>	29	21	carnivorous	37
<i>Carabus germarii</i>	4	3	carnivorous	29.5
<i>Carabus granulatus</i>	671	164	carnivorous	21.5
<i>Carabus hortensis</i>	5	4	carnivorous	26
<i>Carabus intricatus</i>	4	3	carnivorous	31

<i>Carabus monilis</i>	375	45	carnivorous	24
<i>Carabus nemoralis</i>	102	53	carnivorous	23
<i>Carabus problematicus</i>	1	1	carnivorous	25
<i>Carabus purpurascens</i>	41	25	carnivorous	30
<i>Carabus ulrichii</i>	7	5	carnivorous	24.5
<i>Carabus violaceus</i>	159	158	carnivorous	30
<i>Carterus fulvipes</i>	4	3	granivorous	8
<i>Chlaenius chrysocephalus</i>	299	86	carnivorous	8.5
<i>Chlaenius decipiens</i>	7	4	carnivorous	11.5
<i>Chlaenius festivus</i>	34	6	carnivorous	15
<i>Chlaenius nigricornis</i>	9	7	carnivorous	11
<i>Chlaenius olivieri</i>	3	2	carnivorous	11
<i>Chlaenius tristis</i>	1	1	carnivorous	11.5
<i>Cicindela campestris</i>	4	3	carnivorous	12
<i>Clivina collaris</i>	16	11	carnivorous	4.5
<i>Clivina fossor</i>	1491	201	carnivorous	6
<i>Cychrus caraboides</i>	2	2	carnivorous	17
<i>Cylindera germanica</i>	3	2	carnivorous	9.5
<i>Cylindera paludosa</i>	1	1	carnivorous	10.5
<i>Demetrias atricapillus</i>	302	175	carnivorous	5
<i>Diachromus germanus</i>	58	31	granivorous	8.5
<i>Dixus capito</i>	26	3	granivorous	12.5
<i>Dixus clypeatus</i>	6	6	granivorous	10
<i>Dixus sphaerocephalus</i>	1	1	granivorous	6.5
<i>Dolichus halensis</i>	22	6	carnivorous	16.5
<i>Dromius quadrimaculatus</i>	56	96	carnivorous	5
<i>Dromius schneideri</i>	1	1	carnivorous	5.5
<i>Drypta dentata</i>	4	4	carnivorous	8
<i>Dyschirius globosus</i>	50	15	carnivorous	2.5
<i>Elaphrus uliginosus</i>	1	1	carnivorous	9
<i>Epaphius secalis</i>	416	38	carnivorous	3.5
<i>Gynandromorphus etruscus</i>	14	13	granivorous	10.5
<i>Harpalus affinis</i>	9406	909	granivorous	10
<i>Harpalus atratus</i>	76	21	granivorous	11.5
<i>Harpalus attenuatus</i>	3	3	granivorous	8.5
<i>Harpalus calceatus</i>	5	3	granivorous	12.5
<i>Harpalus cupreus</i>	18	12	granivorous	12.5
<i>Harpalus dimidiatus</i>	1272	374	granivorous	12.5
<i>Harpalus distinguendus</i>	1725	376	granivorous	9
<i>Harpalus flavescens</i>	2	2	granivorous	12
<i>Harpalus griseus</i>	18	12	omnivorous	10
<i>Harpalus honestus</i>	4	3	granivorous	8.5
<i>Harpalus laevipes</i>	5	4	granivorous	10.5
<i>Harpalus latus</i>	152	107	granivorous	9.5
<i>Harpalus luteicornis</i>	267	111	granivorous	6.5
<i>Harpalus marginellus</i>	1	1	granivorous	10
<i>Harpalus oblitus</i>	40	31	granivorous	9.5
<i>Harpalus politus</i>	1	1	granivorous	9.5
<i>Harpalus pumilus</i>	1	1	granivorous	5
<i>Harpalus rubripes</i>	154	98	granivorous	10

<i>Harpalus rufipes</i>	10163	1242	granivorous	13.5
<i>Harpalus serripes</i>	61	18	granivorous	10.5
<i>Harpalus signaticornis</i>	716	141	granivorous	6.5
<i>Harpalus smaragdinus</i>	2	1	granivorous	9.5
<i>Harpalus subcylindricus</i>	10	7	granivorous	6.5
<i>Harpalus sulphuripes</i>	3	3	granivorous	7
<i>Harpalus tardus</i>	180	92	carnivorous	9
<i>Harpalus tenebrosus</i>	2	1	granivorous	9
<i>Laemostenus terricola</i>	1	1	carnivorous	14
<i>Leistus ferrugineus</i>	78	37	carnivorous	6.5
<i>Leistus fulvibarbis</i>	3	3	carnivorous	7.5
<i>Leistus rufomarginatus</i>	7	7	carnivorous	8
<i>Leistus spinibarbis</i>	22	12	carnivorous	8.5
<i>Licinus depressus</i>	2	2	carnivorous	10
<i>Licinus punctatulus</i>	1	1	carnivorous	15
<i>Limodromus assimilis</i>	465	147	carnivorous	11.5
<i>Loricera pilicornis</i>	3076	767	carnivorous	7
<i>Microlestes luctuosus</i>	8	5	carnivorous	2
<i>Microlestes maurus</i>	309	119	carnivorous	2
<i>Microlestes minutulus</i>	265	123	carnivorous	2.5
<i>Microlestes negrita</i>	3	3	carnivorous	2
<i>Molops elatus</i>	33	6	carnivorous	15
<i>Molops piceus</i>	16	16	carnivorous	11.5
<i>Nebria brevicollis</i>	1517	461	carnivorous	11.5
<i>Nebria salina</i>	5303	547	carnivorous	11
<i>Notiophilus aestuans</i>	334	137	carnivorous	4.5
<i>Notiophilus aquaticus</i>	6	6	carnivorous	5
<i>Notiophilus biguttatus</i>	2315	361	carnivorous	4.5
<i>Notiophilus germinyi</i>	63	34	carnivorous	5
<i>Notiophilus palustris</i>	235	135	carnivorous	5
<i>Notiophilus quadripunctatus</i>	150	112	carnivorous	4
<i>Notiophilus rufipes</i>	4	4	carnivorous	5
<i>Notiophilus substriatus</i>	32	31	carnivorous	4.5
<i>Ocys harpaloides</i>	2	97	carnivorous	5
<i>Olisthopus rotundatus</i>	1	1	carnivorous	6.5
<i>Ophonus ardosiacus</i>	148	40	granivorous	12
<i>Ophonus azureus</i>	310	137	granivorous	7.5
<i>Ophonus diffinis</i>	2	2	granivorous	11
<i>Ophonus laticollis</i>	52	21	granivorous	9.5
<i>Ophonus puncticeps</i>	19	16	granivorous	8
<i>Ophonus puncticollis</i>	1	1	granivorous	8
<i>Ophonus rufibarbis</i>	23	14	granivorous	7.5
<i>Ophonus rupicola</i>	4	4	granivorous	8
<i>Ophonus sabulicola</i>	4	4	granivorous	15
<i>Ophonus schaubergerianus</i>	26	12	granivorous	8.5
<i>Ophonus subquadratus</i>	3	2	granivorous	7
<i>Orthomus expansus</i>	7	6	carnivorous	10
<i>Oxypselaphus obscurus</i>	9	7	carnivorous	5
<i>Panagaeus bipustulatus</i>	24	15	omnivorous	7
<i>Panagaeus cruxmajor</i>	2	1	omnivorous	8

<i>Paradromius linearis</i>	4	3	carnivorous	4.5
<i>Paranchus albipes</i>	3	3	carnivorous	7.5
<i>Parophonus maculicornis</i>	11	9	granivorous	6
<i>Parophonus mendax</i>	16	16	granivorous	8
<i>Patrobis atrorufus</i>	40	8	carnivorous	8.5
<i>Pedius longicollis</i>	74	27	carnivorous	6
<i>Philorhizus melanocephalus</i>	1	1	carnivorous	2.5
<i>Philorhizus notatus</i>	2	2	carnivorous	2.5
<i>Poecilus cupreus</i>	65826	1651	omnivorous	11
<i>Poecilus gisellae</i>	1	1	carnivorous	9.5
<i>Poecilus lepidus</i>	45	22	carnivorous	12
<i>Poecilus puncticollis</i>	3	2	carnivorous	9
<i>Poecilus purpurascens</i>	27	21	carnivorous	10
<i>Poecilus versicolor</i>	1000	147	carnivorous	10
<i>Polistichus connexus</i>	6	6	carnivorous	8
<i>Porotachys bisulcatus</i>	1	1	carnivorous	2.5
<i>Pterostichus anthracinus</i>	2976	48	carnivorous	10.5
<i>Pterostichus burmeisteri</i>	12	12	carnivorous	13.5
<i>Pterostichus diligens</i>	2	2	carnivorous	5
<i>Pterostichus macer</i>	160	53	carnivorous	13
<i>Pterostichus madidus</i>	3872	325	carnivorous	16.5
<i>Pterostichus melanarius</i>	37257	1330	carnivorous	15
<i>Pterostichus melas</i>	30	9	carnivorous	16
<i>Pterostichus niger</i>	764	127	carnivorous	18.5
<i>Pterostichus nigrita</i>	40	105	carnivorous	10
<i>Pterostichus oblongopunctatus</i>	35	25	carnivorous	10.5
<i>Pterostichus ovoideus</i>	33	16	carnivorous	7
<i>Pterostichus pumilio</i>	1	1	carnivorous	5
<i>Pterostichus strenuus</i>	243	141	carnivorous	6
<i>Pterostichus vernalis</i>	1623	338	carnivorous	6.5
<i>Scybalicus oblongiusculus</i>	16	4	granivorous	11.5
<i>Sinechostictus stomoides</i>	2	2	carnivorous	5.5
<i>Stenolophus marginatus</i>	1	1	omnivorous	6.5
<i>Stenolophus mixtus</i>	1	1	carnivorous	5.5
<i>Stenolophus teutonius</i>	12	9	omnivorous	6
<i>Stomis benoiti</i>	1	1	carnivorous	10
<i>Stomis pumicatus</i>	169	86	carnivorous	7
<i>Syntomus foveatus</i>	33	22	carnivorous	2.5
<i>Syntomus obscuroguttatus</i>	166	88	carnivorous	2.5
<i>Syntomus truncatellus</i>	43	29	carnivorous	2.5
<i>Synuchus vivalis</i>	89	55	granivorous	7
<i>Tachys bistriatus</i>	90	28	carnivorous	1.5
<i>Trechoblemus micros</i>	28	10	carnivorous	4.5
<i>Trechus obtusus</i>	10	105	carnivorous	3.5
<i>Trechus quadristriatus</i>	16780	818	carnivorous	3.5
<i>Trechus rubens</i>	23	15	carnivorous	5.5
<i>Trichotichnus laevicollis</i>	3	3	granivorous	7
<i>Trichotichnus nitens</i>	1	1	granivorous	8
<i>Zabrus tenebrioides</i>	141	62	granivorous	15
<i>Zuphium olens</i>	2	2	carnivorous	9.25

Table S6 Carabids according to diet and body-size

Table S6: Distribution of diet preferences and body-size classes among all carabid beetles identified to species level across all datasets, sites and plots.														
	predominantly carnivorous species				omnivorous species				predominantly granivorous species				overall	
	species richness		individuals		species richness		individuals		species richness		individuals		species richness	individuals
	number	percentage	number	percentage	number	percentage	number	percentage	number	percentage	number	percentage		
small species (< 6 mm)	57	23.65	52,833	15.37	7	2.90	378	0.11	4	1.66	8	< 0.01	68	53,219
medium-sized species (6 - 10 mm)	50	20.75	114,245	33.25	5	2.07	57	0.02	50	20.75	36,813	10.71	105	151,115
large species (> 10 mm)	46	19.09	60,526	17.61	2	0.83	65,872	19.17	20	8.30	12,902	3.75	68	139,300
overall	153	63.49	227,604	66.23	14	5.81	66,307	19.30	74	30.71	49,723	14.47	241	343,634

Table S7 Model specifications for the calculated models

Table S7: Model specifications for the calculated models.			
response	constant effect	varying effect	family
<i>richness (overall)</i>	distance log(trap days) crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Poisson (log)
<i>richness (predominantly carnivorous)</i>	distance log(trap days) crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Poisson (log)
<i>richness (omnivorous)</i> [models with and without <i>P. cupreus</i>]	distance log(trap days) crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Poisson (log)
<i>richness (predominantly granivorous)</i>	distance log(trap days) crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Poisson (log)
<i>activity density (overall)</i>	distance crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Gamma (log)
<i>activity density (predominantly carnivorous)</i>	distance crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Gamma (log)
<i>activity density (omnivorous)</i> [models with and without <i>P. cupreus</i>]	distance crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Gamma (log)
<i>activity density (predominantly granivorous)</i>	distance crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Gamma (log)
<i>activity density small species (< 6 mm)</i>	distance crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Gamma (log)
<i>activity density medium-sized species (6 - 10 mm)</i>	distance crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Gamma (log)
<i>activity density large species (> 10 mm)</i>	distance crop type adjacent habitat distance : crop type distance : adjacent habitat	1 + distance study/site	Gamma (log)

Table S8 Conditional model outputs

Table S8: Conditional model outputs for each response variable tested with estimate (median of the posterior sample), error (standard deviation of posterior sample), lower bound of the 95% credible interval (CI) and upper bound of the 95% credible interval (CI) for each of the constant (fixed) and varying (random) effects used.

reponse		parameter	estimate	95% CI lower bound	95% CI upper bound		
richness (overall)							
constant	slope	adj.control	0.973	0.947	0.998		
		adj.herb	0.971	0.954	0.985		
		adj.woody	0.961	0.937	0.984		
		crop.cereal	0.973	0.947	0.998		
		crop.legume	0.987	0.940	1.036		
		crop.maize	0.910	0.857	0.967		
		crop.oilseed	0.967	0.940	0.997		
		crop.vegetable	0.931	0.870	0.997		
		log.trap.days	1.048	1.033	1.063		
		intercept	adj.control	1.740	1.037	3.016	
	adj.herb		1.659	0.991	2.872		
	adj.woody		1.601	0.954	2.778		
	crop.cereal		1.740	1.037	3.016		
	crop.legume		1.606	0.937	2.796		
	crop.maize		1.481	0.871	2.579		
	crop.oilseed		2.185	1.299	3.829		
	crop.vegetable		1.728	1.005	3.014		
	varying		study	cor.intercept:distance	0.406	-0.452	0.963
				sd.distance	0.167	0.046	0.333
				sd.intercept	0.535	0.398	0.732
		study:site	cor.intercept:distance	-0.297	-0.972	0.869	
sd.distance			0.048	0.002	0.133		
sd.intercept			0.245	0.219	0.272		
summary			R2	0.863	0.856	0.870	
richness (predominantly carnivorous)							
constant	slope	adj.control	0.991	0.961	1.022		
		adj.herb	0.980	0.962	0.993		
		adj.woody	0.963	0.936	0.987		
		crop.cereal	0.991	0.961	1.022		
		crop.legume	1.008	0.951	1.070		
		crop.maize	0.901	0.841	0.965		
		crop.oilseed	0.977	0.945	1.011		
		crop.vegetable	0.975	0.902	1.056		
		log.trap.days	1.039	1.022	1.055		
		intercept	adj.control	1.536	0.871	2.801	
	adj.herb		1.518	0.862	2.750		
	adj.woody		1.528	0.866	2.793		
	crop.cereal		1.536	0.871	2.801		
	crop.legume		1.209	0.668	2.234		
	crop.maize		1.222	0.686	2.248		
	crop.oilseed		1.626	0.912	2.973		
	crop.vegetable		1.463	0.805	2.782		
	varying		study	cor.intercept:distance	0.352	-0.733	0.969
				sd.distance	0.120	0.007	0.304
				sd.intercept	0.600	0.439	0.820
		study:site	cor.intercept:distance	-0.039	-0.944	0.938	
sd.distance			0.045	0.002	0.126		
sd.intercept			0.209	0.178	0.242		
summary			R2	0.808	0.796	0.819	

richness (omnivorous)

constant	slope	adj.control	1.042	0.977	1.108
		adj.herb	1.015	0.985	1.053
		adj.woody	1.042	0.987	1.101
		crop.cereal	1.042	0.977	1.108
		crop.legume	1.020	0.919	1.130
		crop.maize	1.131	1.000	1.275
		crop.oilseed	1.013	0.940	1.088
		crop.vegetable	1.015	0.874	1.178
		log.trap.days	1.049	1.021	1.075
	intercept	adj.control	0.133	0.053	0.361
		adj.herb	0.133	0.054	0.358
		adj.woody	0.126	0.050	0.345
		crop.cereal	0.133	0.053	0.361
		crop.legume	0.163	0.062	0.458
		crop.maize	0.122	0.047	0.342
		crop.oilseed	0.164	0.064	0.454
		crop.vegetable	0.069	0.022	0.234
varying	study	cor.intercept:distance	-0.178	-0.979	0.912
		sd.distance	0.256	0.011	0.745
		sd.intercept	0.826	0.572	1.182
	study:site	cor.intercept:distance	-0.162	-0.971	0.920
		sd.distance	0.065	0.003	0.187
		sd.intercept	0.027	0.001	0.077
summary		R2	0.450	0.416	0.481

richness (omnivorous – without P. cupreus)

constant	slope	adj.control	0.975	0.888	1.066
		adj.herb	0.966	0.896	1.035
		adj.woody	0.988	0.870	1.114
		crop.cereal	0.975	0.888	1.066
		crop.legume	0.955	0.809	1.122
		crop.maize	1.012	0.862	1.193
		crop.oilseed	0.908	0.794	1.035
		crop.vegetable	0.954	0.809	1.126
		log.trap.days	1.112	1.066	1.155
	intercept	adj.control	0.002	0.000	0.012
		adj.herb	0.002	0.000	0.012
		adj.woody	0.002	0.001	0.012
		crop.cereal	0.002	0.000	0.012
		crop.legume	0.002	0.000	0.016
		crop.maize	0.008	0.002	0.049
		crop.oilseed	0.003	0.001	0.017
		crop.vegetable	0.003	0.000	0.020
varying	study	cor.intercept:distance	0.064	-0.939	0.962
		sd.distance	0.364	0.013	1.117
		sd.intercept	1.059	0.609	1.794
	study:site	cor.intercept:distance	-0.032	-0.954	0.945
		sd.distance	0.339	0.012	0.943
		sd.intercept	0.299	0.021	0.618
summary		R2	0.263	0.196	0.340

richness (predominantly granivorous)

constant	slope	adj.control	0.917	0.870	0.968
		adj.herb	0.915	0.874	0.951
		adj.woody	0.912	0.859	0.971
		crop.cereal	0.917	0.870	0.968
		crop.legume	0.953	0.873	1.034
		crop.maize	0.870	0.775	0.980
		crop.oilseed	0.955	0.899	1.018
		crop.vegetable	0.879	0.782	0.988
		log.trap.days	1.069	1.050	1.088
	intercept	adj.control	0.219	0.113	0.430
		adj.herb	0.198	0.103	0.387
		adj.woody	0.174	0.090	0.339
		crop.cereal	0.219	0.113	0.430
		crop.legume	0.253	0.128	0.502
		crop.maize	0.216	0.107	0.434
		crop.oilseed	0.406	0.212	0.806
		crop.vegetable	0.242	0.117	0.511
varying	study	cor.intercept:distance	0.189	-0.408	0.728
		sd.distance	0.714	0.371	1.151
		sd.intercept	0.518	0.372	0.721
	study:site	cor.intercept:distance	0.341	-0.774	0.978
		sd.distance	0.117	0.005	0.283
		sd.intercept	0.299	0.245	0.352
summary		R2	0.755	0.737	0.771

activity density (overall)

constant	slope	adj.control	1.037	0.986	1.085
		adj.herb	0.996	0.971	1.021
		adj.woody	1.035	0.995	1.078
		crop.cereal	1.037	0.986	1.085
		crop.legume	1.063	0.979	1.152
		crop.maize	0.937	0.858	1.020
		crop.oilseed	1.036	0.976	1.094
		crop.vegetable	1.007	0.917	1.108
	intercept	adj.control	1.845	1.181	2.788
		adj.herb	1.810	1.187	2.703
		adj.woody	1.762	1.153	2.717
		crop.cereal	1.845	1.181	2.788
		crop.legume	1.664	1.007	2.633
		crop.maize	1.779	1.087	2.845
		crop.oilseed	2.228	1.389	3.429
		crop.vegetable	2.406	1.432	4.080
varying	study	cor.intercept:distance	0.172	-0.646	0.810
		sd.distance	0.300	0.029	0.576
		sd.intercept	1.053	0.799	1.400
	study:site	cor.intercept:distance	-0.290	-0.964	0.825
		sd.distance	0.101	0.004	0.284
		sd.intercept	0.747	0.704	0.791
summary		R2	0.550	0.476	0.635

activity density (predominantly carnivorous)

constant	slope	adj.control	1.027	0.975	1.077
		adj.herb	0.988	0.959	1.015
		adj.woody	0.983	0.941	1.026
		crop.cereal	1.027	0.975	1.077
		crop.legume	1.041	0.953	1.134
		crop.maize	0.941	0.859	1.032
		crop.oilseed	1.036	0.975	1.098
		crop.vegetable	1.021	0.922	1.129
	intercept	adj.control	1.206	0.800	1.821
		adj.herb	1.262	0.855	1.871
		adj.woody	1.285	0.843	1.950
		crop.cereal	1.206	0.800	1.821
		crop.legume	0.975	0.614	1.581
		crop.maize	1.064	0.658	1.718
		crop.oilseed	1.222	0.796	1.902
		crop.vegetable	1.534	0.914	2.570
varying	study	cor.intercept:distance	0.406	-0.383	0.929
		sd.distance	0.360	0.049	0.702
		sd.intercept	1.028	0.772	1.375
	study:site	cor.intercept:distance	-0.211	-0.956	0.876
		sd.distance	0.096	0.004	0.264
		sd.intercept	0.730	0.684	0.777
	summary		R2	0.423	0.531

activity density (omnivorous)

constant	slope	adj.control	1.083	1.008	1.161
		adj.herb	1.056	0.999	1.121
		adj.woody	1.171	1.079	1.264
		crop.cereal	1.083	1.008	1.161
		crop.legume	1.121	1.007	1.251
		crop.maize	1.027	0.906	1.168
		crop.oilseed	1.117	1.020	1.220
		crop.vegetable	1.095	0.941	1.279
	intercept	adj.control	0.410	0.258	0.659
		adj.herb	0.421	0.274	0.648
		adj.woody	0.298	0.185	0.482
		crop.cereal	0.410	0.258	0.659
		crop.legume	0.413	0.239	0.728
		crop.maize	0.291	0.167	0.515
		crop.oilseed	0.385	0.233	0.637
		crop.vegetable	0.394	0.173	0.889
varying	study	cor.intercept:distance	-0.299	-0.719	0.207
		sd.distance	0.998	0.627	1.505
		sd.intercept	1.064	0.777	1.449
	study:site	cor.intercept:distance	-0.202	-0.460	0.153
		sd.distance	0.729	0.377	1.070
		sd.intercept	0.815	0.746	0.887
	summary		R2	0.636	0.710

activity density (omnivorous – without P. cupreus)

constant	slope	adj.control	1.026	0.944	1.120
		adj.herb	0.981	0.925	1.044
		adj.woody	1.029	0.920	1.149
		crop.cereal	1.026	0.944	1.120
		crop.legume	1.022	0.873	1.200
		crop.maize	1.042	0.898	1.212
		crop.oilseed	1.026	0.906	1.168
		crop.vegetable	0.988	0.841	1.161
	intercept	adj.control	0.062	0.036	0.106
		adj.herb	0.047	0.030	0.080
		adj.woody	0.053	0.029	0.103
		crop.cereal	0.062	0.036	0.106
		crop.legume	0.060	0.028	0.129
		crop.maize	0.077	0.037	0.161
		crop.oilseed	0.051	0.027	0.102
		crop.vegetable	0.061	0.026	0.148
varying	study	cor.intercept:distance	0.297	-0.782	0.972
		sd.distance	0.376	0.017	1.020
		sd.intercept	0.807	0.531	1.223
	study:site	cor.intercept:distance	-0.251	-0.960	0.874
		sd.distance	0.297	0.011	0.881
		sd.intercept	0.478	0.350	0.613
summary		R2	0.579	0.502	0.656

activity density (predominantly granivorous)

constant	slope	adj.control	0.931	0.890	0.976
		adj.herb	0.964	0.939	0.986
		adj.woody	0.954	0.913	0.998
		crop.cereal	0.931	0.890	0.976
		crop.legume	0.963	0.889	1.046
		crop.maize	0.890	0.805	0.984
		crop.oilseed	0.946	0.898	1.002
		crop.vegetable	0.869	0.781	0.970
	intercept	adj.control	0.283	0.214	0.377
		adj.herb	0.207	0.162	0.266
		adj.woody	0.208	0.158	0.278
		crop.cereal	0.283	0.214	0.377
		crop.legume	0.286	0.198	0.414
		crop.maize	0.402	0.273	0.593
		crop.oilseed	0.779	0.569	1.059
		crop.vegetable	0.455	0.290	0.715
varying	study	cor.intercept:distance	-0.098	-0.917	0.879
		sd.distance	0.133	0.005	0.441
		sd.intercept	0.573	0.405	0.790
	study:site	cor.intercept:distance	0.336	-0.752	0.970
		sd.distance	0.147	0.006	0.372
		sd.intercept	0.736	0.686	0.785
summary	R2	0.429	0.371	0.495	

activity density of small species (< 6 mm)

constant	slope	adj.control	0.973	0.914	1.029
		adj.herb	0.966	0.926	1.000
		adj.woody	0.959	0.903	1.012
		crop.cereal	0.973	0.914	1.029
		crop.legume	0.979	0.884	1.085
		crop.maize	0.877	0.789	0.972
		crop.oilseed	0.998	0.928	1.072
		crop.vegetable	0.965	0.867	1.076
	intercept	adj.control	0.258	0.188	0.365
		adj.herb	0.263	0.198	0.353
		adj.woody	0.313	0.226	0.435
		crop.cereal	0.258	0.188	0.365
		crop.legume	0.243	0.161	0.378
		crop.maize	0.314	0.204	0.480
		crop.oilseed	0.288	0.203	0.414
		crop.vegetable	0.509	0.315	0.820
varying	study	cor.intercept:distance	0.284	-0.317	0.799
		sd.distance	0.532	0.160	0.962
		sd.intercept	0.711	0.525	0.974
	study:site	cor.intercept:distance	0.158	-0.850	0.949
		sd.distance	0.156	0.006	0.427
		sd.intercept	0.723	0.669	0.777
summary	R2	0.188	0.116	0.298	

activity density of medium-sized species (6 – 10 mm)

constant	slope	adj.control	0.984	0.938	1.029
		adj.herb	0.969	0.941	0.993
		adj.woody	1.019	0.977	1.068
		crop.cereal	0.984	0.938	1.029
		crop.legume	0.939	0.862	1.020
		crop.maize	0.993	0.897	1.096
		crop.oilseed	0.983	0.931	1.037
		crop.vegetable	0.941	0.842	1.055
	intercept	adj.control	0.711	0.479	1.064
		adj.herb	0.686	0.468	1.015
		adj.woody	0.593	0.401	0.902
		crop.cereal	0.711	0.479	1.064
		crop.legume	0.617	0.389	1.000
		crop.maize	0.381	0.239	0.617
		crop.oilseed	1.231	0.816	1.902
		crop.vegetable	0.617	0.360	1.058
varying	study	cor.intercept:distance	0.274	-0.686	0.937
		sd.distance	0.242	0.014	0.594
		sd.intercept	1.001	0.751	1.338
	study:site	cor.intercept:distance	0.237	-0.838	0.957
		sd.distance	0.114	0.005	0.315
		sd.intercept	0.675	0.627	0.722
summary	R2	0.410	0.339	0.500	

activity density of large species (> 10 mm)

constant	slope	adj.control	1.078	1.015	1.144
		adj.herb	1.029	0.982	1.078
		adj.woody	1.064	1.003	1.131
		crop.cereal	1.078	1.015	1.144
		crop.legume	1.112	1.013	1.222
		crop.maize	0.998	0.897	1.106
		crop.oilseed	1.101	1.021	1.184
		crop.vegetable	1.035	0.923	1.159
	intercept	adj.control	0.888	0.581	1.362
		adj.herb	0.834	0.561	1.253
		adj.woody	0.788	0.521	1.201
		crop.cereal	0.888	0.581	1.362
		crop.legume	0.829	0.510	1.359
		crop.maize	0.916	0.565	1.544
		crop.oilseed	0.849	0.547	1.329
		crop.vegetable	1.059	0.609	1.806
varying	study	cor.intercept:distance	-0.066	-0.523	0.403
		sd.distance	0.867	0.573	1.250
		sd.intercept	1.023	0.767	1.374
	study:site	cor.intercept:distance	-0.259	-0.569	0.209
		sd.distance	0.542	0.082	0.874
		sd.intercept	0.826	0.770	0.883
summary		R2	0.587	0.491	0.691

Table S9 Marginal estimates for distance slopes

Table S9: Marginal estimates for distance slopes with median, 95% credible interval (CI), percent change in the response per 10 m within-field distance and the probability of the slope being positive and negative calculated from the posterior sample across all crops and adjacent habitat types (translated to percent). Probabilities above 95% are highlighted in bold.

model	crop / habitat	median estimate	95% CI	median change per 10 m [%]	95% CI change per 10 m [%]	probability of positive slope [%]	probability of negative slope [%]
<i>richness (overall)</i>							
	overall	0.958	[0.870; 1.011]	-4.20	[-13.02; 1.07]	5.073	94.927
	cereal	0.968	[0.952; 0.983]	-3.16	[-4.80; -1.69]	0	100
	legume	0.982	[0.940; 1.026]	-1.80	[-6.04; 2.55]	20.85	79.150
	maize	0.905	[0.857; 0.957]	-9.50	[-14.35; -4.25]	0.025	99.975
	oilseed	0.963	[0.942; 0.984]	-3.70	[-5.77; -1.56]	0.050	99.95
	vegetable	0.926	[0.868; 0.989]	-7.39	[-13.21; -1.09]	1.125	98.875
	control	0.953	[0.924; 0.982]	-4.70	[-7.58; -1.77]	0.075	99.925
	herbaceous	0.951	[0.928; 0.973]	-4.93	[-7.18; -2.74]	0	100
	woody	0.942	[0.917; 0.967]	-5.84	[-8.30; -3.30]	0	100
<i>richness (carnivorous)</i>							
	overall	0.966	[0.852; 1.036]	-3.36	[-14.8; 3.6]	14.538	85.463
	cereal	0.978	[0.960; 0.993]	-2.24	[-4.02; -0.71]	0.225	99.775
	legume	0.995	[0.944; 1.048]	-0.51	[-5.64; 4.81]	42.575	57.425
	maize	0.888	[0.835; 0.946]	-11.16	[-16.53; -5.42]	0	100
	oilseed	0.964	[0.941; 0.988]	-3.57	[-5.85; -1.19]	0.3125	99.688
	vegetable	0.962	[0.893; 1.036]	-3.79	[-10.65; 3.62]	15.425	84.575
	control	0.967	[0.936; 1.004]	-3.03	[-6.43; 0.43]	4.05	95.95
	herbaceous	0.958	[0.934; 0.983]	-4.17	[-6.58; -1.72]	0.075	99.925
	woody	0.942	[0.915; 0.969]	-5.80	[-8.54; -3.07]	0.0125	99.988
<i>richness (omnivorous)</i>							
	overall	1.027	[0.913; 1.204]	2.72	[-8.67; 20.42]	70.471	29.529
	cereal	1.033	[0.998; 1.071]	3.30	[-0.20; 7.10]	96.588	3.413
	legume	1.012	[0.924; 1.104]	1.25	[-7.60; 10.40]	60.638	39.363
	maize	1.121	[1.004; 1.249]	12.09	[0.40; 24.90]	97.950	2.050
	oilseed	1.005	[0.950; 1.060]	0.49	[-5.00; 6.00]	57.188	42.813
	vegetable	1.006	[0.878; 1.157]	0.64	[-12.20; 15.70]	53.850	46.150
	control	1.043	[0.970; 1.119]	4.26	[-3.00; 11.90]	88.050	11.950
	herbaceous	1.017	[0.969; 1.069]	1.71	[-3.10; 6.90]	75.338	24.663
	woody	1.044	[0.980; 1.110]	4.37	[-2.00; 11.00]	91.000	9.000
<i>richness (omnivorous – without P. cupreus)</i>							
	overall	0.961	[0.813; 1.143]	-3.89	[-18.60; 14.26]	30.789	69.211
	cereal	0.976	[0.906; 1.046]	-2.39	[-9.40; 4.59]	25.163	74.839
	legume	0.956	[0.819; 1.112]	-4.43	[-18.10; 11.23]	27.538	72.463
	maize	1.013	[0.871; 1.184]	1.34	[-12.90; 18.42]	57.088	42.913
	oilseed	0.909	[0.804; 1.022]	-9.11	[-19.60; 2.23]	5.638	94.363
	vegetable	0.956	[0.815; 1.119]	-4.41	[-18.50; 11.94]	29.238	70.763
	control	0.960	[0.867; 1.059]	-3.96	[-13.30; 5.87]	22.175	77.825
	herbaceous	0.951	[0.872; 1.037]	-4.86	[-12.80; 3.69]	11.725	88.275
	woody	0.973	[0.853; 1.105]	-2.66	[-14.70; 10.54]	33.500	66.500
<i>richness (granivorous)</i>							
	overall	0.918	[0.797; 1.009]	-8.16	[-20.33; 0.91]	3.905	96.095
	cereal	0.914	[0.878; 0.951]	-8.55	[-12.21; -4.86]	0	100
	legume	0.950	[0.878; 1.025]	-4.97	[-12.20; 2.49]	9.025	90.975
	maize	0.867	[0.778; 0.97]	-13.26	[-22.24; -2.96]	0.525	99.475
	oilseed	0.953	[0.907; 1.004]	-4.71	[-9.30; 0.44]	3.450	96.550
	vegetable	0.876	[0.786; 0.981]	-12.39	[-21.39; -1.90]	1.025	98.975
	control	0.913	[0.862; 0.971]	-8.66	[-13.83; -2.88]	0.175	99.825
	herbaceous	0.912	[0.866; 0.955]	-8.84	[-13.44; -4.53]	0.075	99.925
	woody	0.909	[0.855; 0.974]	-9.06	[-14.49; -2.61]	0.425	99.575
<i>activity density (overall)</i>							
	overall	1.008	[0.876; 1.107]	0.84	[-12.37; 10.67]	56.588	43.412
	cereal	1.022	[0.995; 1.048]	2.25	[-0.47; 4.83]	94.950	5.050
	legume	1.047	[0.981; 1.123]	4.75	[-1.95; 12.35]	91.025	8.975
	maize	0.925	[0.857; 0.996]	-7.55	[-14.34; -0.40]	1.975	98.025
	oilseed	1.021	[0.98; 1.064]	2.13	[-1.99; 6.38]	84.250	15.750
	vegetable	0.993	[0.911; 1.086]	-0.71	[-8.94; 8.61]	43.475	56.525
	control	1.015	[0.961; 1.067]	1.53	[-3.86; 6.67]	70.350	29.650
	herbaceous	0.975	[0.942; 1.01]	-2.47	[-5.76; 0.98]	8.000	92.000
	woody	1.014	[0.971; 1.058]	1.38	[-2.90; 5.85]	72.500	27.500

<i>activity density (carnivorous)</i>						
overall	0.991	[0.866; 1.089]	-0.92	[-13.40; 8.90]	41.435	58.565
cereal	0.999	[0.969; 1.027]	-0.06	[-3.10; 2.70]	48.500	51.500
legume	1.013	[0.938; 1.094]	1.28	[-6.22; 9.36]	63.000	37.000
maize	0.916	[0.845; 0.995]	-8.37	[-15.55; -0.52]	1.900	98.100
oilseed	1.008	[0.964; 1.055]	0.79	[-3.64; 5.53]	63.438	36.563
vegetable	0.993	[0.905; 1.091]	-0.67	[-9.48; 9.10]	44.275	55.725
control	1.012	[0.958; 1.066]	1.23	[-4.18; 6.59]	66.450	33.550
herbaceous	0.975	[0.937; 1.013]	-2.53	[-6.33; 1.26]	9.213	90.788
woody	0.967	[0.926; 1.017]	-3.03	[-7.41; 1.70]	9.288	90.713
<i>activity density (omnivorous)</i>						
overall	1.104	[0.947; 1.293]	10.44	[-5.33; 29.30]	91.087	8.913
cereal	1.102	[1.044; 1.163]	10.24	[4.40; 16.30]	99.963	0.038
legume	1.140	[1.036; 1.259]	14.01	[3.60; 25.90]	99.638	0.363
maize	1.046	[0.928; 1.171]	4.64	[-7.20; 17.10]	77.988	22.013
oilseed	1.136	[1.052; 1.223]	13.64	[5.20; 22.30]	99.950	0.050
vegetable	1.115	[0.961; 1.290]	11.50	[-3.90; 29.00]	92.850	7.150
control	1.089	[1.005; 1.177]	8.86	[0.50; 17.70]	98.188	1.813
herbaceous	1.062	[0.994; 1.136]	6.19	[-0.60; 13.60]	96.313	3.688
woody	1.176	[1.078; 1.276]	17.64	[7.80; 27.60]	99.988	0.013
<i>activity density (omnivorous – without P. cupreus)</i>						
overall	1.006	[0.862; 1.176]	0.58	[-13.80; 17.59]	53.233	46.767
cereal	1.011	[0.951; 1.079]	1.15	[-4.90; 7.90]	63.813	36.188
legume	1.008	[0.868; 1.175]	0.81	[-13.20; 17.52]	54.313	45.688
maize	1.027	[0.898; 1.178]	2.72	[-10.20; 17.79]	65.375	34.625
oilseed	1.012	[0.904; 1.135]	1.20	[-9.60; 13.50]	58.438	41.563
vegetable	0.975	[0.839; 1.133]	-2.50	[-16.10; 13.35]	36.513	63.488
control	1.020	[0.930; 1.125]	2.04	[-7.00; 12.52]	66.475	33.525
herbaceous	0.976	[0.906; 1.055]	-2.43	[-9.40; 5.53]	26.438	73.563
woody	1.024	[0.912; 1.153]	2.39	[-8.80; 15.26]	66.038	33.963
<i>activity density (granivorous)</i>						
overall	0.946	[0.825; 1.035]	-5.41	[-17.55; 3.47]	9.960	90.040
cereal	0.950	[0.925; 0.975]	-5.04	[-7.51; -2.48]	0.025	99.975
legume	0.983	[0.917; 1.059]	-1.74	[-8.35; 5.93]	32.000	68.000
maize	0.907	[0.827; 0.992]	-9.26	[-17.27; -0.76]	1.725	98.275
oilseed	0.964	[0.926; 1.005]	-3.57	[-7.36; 0.55]	4.125	95.875
vegetable	0.887	[0.802; 0.979]	-11.29	[-19.78; -2.06]	0.950	99.050
control	0.919	[0.875; 0.969]	-8.09	[-12.46; -3.08]	0.100	99.900
herbaceous	0.952	[0.917; 0.987]	-4.82	[-8.31; -1.28]	0.425	99.575
woody	0.942	[0.900; 0.989]	-5.83	[-10.05; -1.09]	0.975	99.025
<i>activity density of small species (< 6 mm)</i>						
overall	0.960	[0.819; 1.056]	-3.98	[-18.07; 5.62]	19.723	80.277
cereal	0.966	[0.927; 1.001]	-3.41	[-7.33; 0.07]	2.800	97.200
legume	0.972	[0.885; 1.069]	-2.80	[-11.49; 6.92]	27.925	72.075
maize	0.871	[0.792; 0.955]	-12.95	[-20.75; -4.46]	0.175	99.825
oilseed	0.990	[0.936; 1.049]	-0.99	[-6.43; 4.89]	36.050	63.950
vegetable	0.958	[0.868; 1.059]	-4.25	[-13.15; 5.94]	19.825	80.175
control	0.957	[0.897; 1.016]	-4.26	[-10.29; 1.56]	8.100	91.900
herbaceous	0.950	[0.904; 0.996]	-5.01	[-9.56; -0.44]	1.450	98.550
woody	0.944	[0.888; 0.999]	-5.60	[-11.23; -0.10]	2.300	97.700
<i>activity density of medium-sized species (6 – 10 mm)</i>						
overall	0.976	[0.871; 1.076]	-2.43	[-12.87; 7.58]	30.218	69.783
cereal	0.991	[0.964; 1.016]	-0.92	[-3.55; 1.57]	23.375	76.625
legume	0.944	[0.876; 1.018]	-5.59	[-12.42; 1.84]	6.963	93.038
maize	0.999	[0.911; 1.095]	-0.08	[-8.89; 9.51]	49.263	50.738
oilseed	0.989	[0.950; 1.033]	-1.09	[-5.04; 3.26]	30.088	69.913
vegetable	0.947	[0.854; 1.055]	-5.26	[-14.58; 5.5]	16.300	83.700
control	0.968	[0.918; 1.019]	-3.22	[-8.20; 1.90]	11.100	88.900
herbaceous	0.952	[0.915; 0.990]	-4.76	[-8.53; -0.95]	0.888	99.113
woody	1.002	[0.957; 1.056]	0.20	[-4.35; 5.62]	53.025	46.975
<i>activity density of large species (> 10 mm)</i>						
overall	1.048	[0.908; 1.168]	4.83	[-9.21; 16.82]	76.333	23.667
cereal	1.057	[1.012; 1.104]	5.69	[1.20; 10.45]	99.340	0.660
legume	1.090	[1.003; 1.185]	8.97	[0.32; 18.55]	97.920	2.080
maize	0.978	[0.887; 1.073]	-2.21	[-11.25; 7.31]	31.750	68.250
oilseed	1.079	[1.014; 1.148]	7.90	[1.38; 14.84]	99.040	0.960
vegetable	1.014	[0.912; 1.130]	1.43	[-8.85; 12.97]	60.430	39.570
control	1.064	[0.996; 1.134]	6.44	[-0.36; 13.37]	96.820	3.180
herbaceous	1.015	[0.962; 1.071]	1.52	[-3.80; 7.10]	71.120	28.880
woody	1.050	[0.985; 1.120]	4.95	[-1.47; 11.95]	93.280	6.720

Supplementary references

- Anjum-Zubair, M., Schmidt-Entling, M.H., Querner, P. & Frank, T. (2010). Influence of within-field position and adjoining habitat on carabid beetle assemblages in winter wheat. *Agricultural and Forest Entomology*, 12, 301-306.
- Badenhausser, I., Gross, N., Mornet, V., Roncoroni, M., Saintilan, A., Rusch, A. (2020). Increasing amount and quality of green infrastructures at different scales promotes biological control in agricultural landscapes. *Agriculture, Ecosystems & Environment*, 290, 106735.
- Batáry, P., Gallé, R., Riesch, F., Fischer, C., Dormann, C.F., Mußhoff, O. et al. (2017). The former Iron Curtain still drives biodiversity–profit trade-offs in German agriculture. *Nature Ecology & Evolution*, 1, 1279-1284.
- Batáry, P., Holzschuh, A., Orci, K.M., Samu, F. & Tscharntke, T. (2012). Responses of plant, insect and spider biodiversity to local and landscape scale management intensity in cereal crops and grasslands. *Agriculture, Ecosystems & Environment*, 146, 130-136.
- Birkhofer, K., Wolters, V. & Diekötter, T. (2014). Grassy margins along organically managed cereal fields foster trait diversity and taxonomic distinctness of arthropod communities. *Insect Conservation and Diversity*, 7, 274-287.
- Boetzl, F.A., Krimmer, E., Krauss, J. & Steffan-Dewenter, I. (2019). Agri-environmental schemes promote ground-dwelling predators in adjacent oilseed rape fields: Diversity, species traits and distance-decay functions. *Journal of Applied Ecology*, 56, 10-20.
- Boetzl, F.A., Schuele, M., Krauss, J. & Steffan-Dewenter, I. (2020). Pest control potential of adjacent agri-environment schemes varies with crop type and is shaped by landscape context and within-field position. *Journal of Applied Ecology*, 57, 1482-1493.
- Cutler, G.C., Renkema, J.M., Majka, C., Sproule, J. (2012). Distribution and abundance of Carabidae in Nova Scotia wild blueberry fields. *The Canadian Entomologist*, 144, 1-13.

- Ditner, N., Balmer, O., Beck, J., Blick, T., Nagel, P. & Luka, H. (2013). Effects of experimentally planting non-crop flowers into cabbage fields on the abundance and diversity of predators. *Biodiversity and Conservation*, 22, 1049-1061.
- French, B.W., & Elliott, N.C. (1999). Spatial and temporal distribution of ground beetle (Coleoptera: Carabidae) assemblages in riparian strips and adjacent wheat fields. *Environmental Entomology*, 28, 597-607.
- French, B.W., Elliott, N.C., Berberet, R.C., & Burd, J.D. (2001). Effects of riparian and grassland habitats on ground beetle (Coleoptera: Carabidae) assemblages in adjacent wheat fields. *Environmental Entomology*, 30, 225-234.
- Fusser, M.S., Holland, J.M., Jeanneret, P., Pfister, S.C., Entling, M.H. & Schirmel, J. (2018). Interactive effects of local and landscape factors on farmland carabids. *Agricultural and Forest Entomology*, 20, 549-557.
- González, E., Salvo A., Valladares, G. (2017). Arthropod communities and biological control in soybean fields: Forest cover at landscape scale is more influential than forest proximity. *Agriculture, Ecosystems & Environment*, 239, 359-367.
- Hanson, H.I., Palmu, E., Birkhofer, K., Smith, H.G. & Hedlund, K. (2016). Agricultural Land Use Determines the Trait Composition of Ground Beetle Communities. *PLoS One*, 11, 13.
- Hof, A.R. & Bright, P.W. (2010). The impact of grassy field margins on macro-invertebrate abundance in adjacent arable fields. *Agriculture, Ecosystems & Environment*, 139, 280-283.
- Homburg, K., Homburg, N., Schäfer, F., Schuldt, A. & Assmann, T. (2014). Carabids.org – a dynamic online database of ground beetle species traits (Coleoptera, Carabidae). *Insect Conservation and Diversity*, 7, 195-205.
- Hůrka, K. (1996). Carabidae of the Czech and Slovak republics. Ing. Vit Kabourek.

- Knapp, M., Seidl, M., Knappová, J., Macek, M. & Saska, P. (2019). Temporal changes in the spatial distribution of carabid beetles around arable field-woodlot boundaries. *Scientific Reports*, 9, 8967.
- Koivula, M., Hyyryläinen, V., & Soininen, E. (2004). Carabid beetles (Coleoptera: Carabidae) at forest-farmland edges in southern Finland. *Journal of Insect Conservation*, 8, 297-309.
- Loureiro A. (2018). Feeding behaviour, dispersal, and community dynamics of Carabidae found in Nova Scotia lowbush blueberry fields. Dalhousie University (Thesis).
- Maas, B., Brandl, M., Hussain, R.I., Frank, T., Zulka, K.P., Rabl, D. et al. (2021). Functional traits driving pollinator and predator responses to newly established grassland strips in agricultural landscapes. *Journal of Applied Ecology*, 58, 1728-1737.
- Madeira, F., Tschardtke, T., Elek, Z., Kormann, U.G., Pons, X., Rösch, V. et al. (2016). Spillover of arthropods from cropland to protected calcareous grassland – the neighbouring habitat matters. *Agriculture, Ecosystems & Environment*, 235, 127-133.
- Marrec, R., Caro, G., Miguet, P., Badenhausser, I., Plantegenest, M., Vialatte, A. et al. (2017). Spatiotemporal dynamics of the agricultural landscape mosaic drives distribution and abundance of dominant carabid beetles. *Landscape Ecology*, 32, 2383-2398.
- Martin, E.A., Seo, B., Park, C.-R., Reineking, B. and Steffan-Dewenter, I. (2016). Scale-dependent effects of landscape composition and configuration on natural enemy diversity, crop herbivory, and yields. *Ecological Applications*, 26, 448-462.
- Nash, M.A., Thomson, L.J., Hoffmann, A.A. (2008). Effect of remnant vegetation, pesticides, and farm management on abundance of the beneficial predator *Notonomus gravis* (Chaudoir) (Coleoptera: Carabidae). *Biological Control*, 46, 83-93.
- Ng, K., Barton, P. S., Macfadyen, S., Lindenmayer, D. B., Driscoll, D. A. (2018). Beetle's responses to edges in fragmented landscapes are driven by adjacent farmland use, season and cross-habitat movement. *Landscape Ecology*, 33, 109-125.

- Pakeman, R.J. & Stockan, J.A. (2014). Drivers of carabid functional diversity: abiotic environment, plant functional traits, or plant functional diversity? *Ecology*, 95, 1213-1224.
- Pfiffner, L. & Luka, H. (2003). Effects of low-input farming systems on carabids and epigeal spiders - a paired farm approach. *Basic and Applied Ecology*, 4, 117-127.
- Ribera, I., Dolédec, S., Downie, I.S. & Foster, G.N. (2001). Effect of land disturbance and stress on species traits of ground beetle assemblages. *Ecology*, 82, 1112-1129.
- Saska P., Vodde M., Heijerman T., Westerman P., van der Werf W. (2007). The significance of a grassy field boundary for the spatial distribution of carabids within two cereal fields. *Agriculture, Ecosystems & Environment*, 122, 427-434.
- Sirami, C., Gross, N., Baillod, A.B., Bertrand, C., Carrié, R., Hass, A. et al. (2019). Increasing crop heterogeneity enhances multitrophic diversity across agricultural regions. *Proceedings of the National Academy of Sciences*, 116, 16442-16447.
- Sutter, L., Albrecht, M., Jeanneret, P. & Diekötter, T. (2018). Landscape greening and local creation of wildflower strips and hedgerows promote multiple ecosystem services. *Journal of Applied Ecology*, 55, 612-620.
- Tschumi, M., Albrecht, M., Entling, M.H. & Jacot, K. (2015). High effectiveness of tailored flower strips in reducing pests and crop plant damage. *Proceedings of the Royal Society B: Biological Sciences*, 282, 189-196.
- Van Vooren, L., Reubens, B., Ampoorter, E., Broekx, S., Pardon, P., Van Waes, C. et al. (2018). Monitoring the Impact of Hedgerows and Grass Strips on the Performance of Multiple Ecosystem Service Indicators. *Environmental Management*, 62, 241-259.